TEXORA · LECTURE SHEET

Moment, Skewness and Kurtosis

Raw and central moments and the relations between them, the meaning and measurement of skewness by the Pearson and Bowley coefficients, and kurtosis as a measure of peakedness through the beta coefficients.

STAT-2109 · PROBABILITY AND STATISTICS · TOPIC 5 · LEVEL 2 TERM 1

1. What a moment is

A mean tells us where a distribution is centred and a standard deviation tells us how widely it spreads, but two distributions can share the same mean and the same standard deviation and still look quite different. One may lean to the left, the other to the right; one may be sharply peaked, the other flat. To describe shape completely we need a systematic family of averages built from powers of the deviations. Those averages are the **moments** of the distribution.

Formally, a moment is the arithmetic mean of a power of a deviation. Two choices of origin give two families. If deviations are taken from an **arbitrary point** A, usually a convenient working origin, we obtain **raw moments about A**, also called moments about an arbitrary origin. If deviations are taken from the actual mean, we obtain **central moments**. Central moments describe the distribution in its own terms, independent of where it sits on the scale, and they are the moments used for skewness and kurtosis.

The order of a moment is the power used. The first moment is built from first powers, the second from squares, the third from cubes, and the fourth from fourth powers. Each order carries distinct information: order one locates, order two spreads, order three tilts, and order four describes the peak and the tails.

2. Raw moments about an arbitrary point

For observations $x_1, x_2, \ldots, x_n$ with frequencies $f_1, f_2, \ldots$ (for ungrouped data every $f = 1$), the **r-th raw moment about an arbitrary point A** is

$$\mu'_r = \Sigma f (x - A)^r / \Sigma f , \quad r = 1, 2, 3, 4, \ldots$$

The prime distinguishes a raw moment from a central moment. The first four raw moments about A are

$$\mu'_1 = \Sigma f (x - A) / N, \quad \mu'_2 = \Sigma f (x - A)^2 / N, \quad \mu'_3 = \Sigma f (x - A)^3 / N, \quad \mu'_4 = \Sigma f (x - A)^4 / N$$

where $N = \Sigma f$. Raw moments are easy to compute in a table because A can be chosen as a round number near the middle of the data, which keeps the deviations $(x - A)$ small. Their weakness is that they depend on the arbitrary choice of A, so they cannot by themselves describe shape. The first raw moment does have a direct use: the mean is recovered from it as $\bar{x} = A + \mu'_1$. Everything else of interest is transferred to central moments by the relations in Section 4.

For grouped data the class mid-values play the role of x, and the class frequencies play the role of f. The formulae are unchanged. This is why grouped tables in this course always begin with a mid-value column: every later summary, from the mean to the fourth moment, is built from those mid-values.

3. Central moments

The **r-th central moment** is the mean of the r-th power of deviations from the arithmetic mean $\bar{x}$:

$$\mu_r = \Sigma f (x - \bar{x})^r / \Sigma f , \quad r = 1, 2, 3, 4, \ldots$$

The first four central moments are interpreted as follows.

MOMENT	DEFINITION	MEANING
μ_1	$\Sigma f (x - \bar{x}) / N$	Always zero, because positive and negative deviations from the mean cancel exactly. It is a check on arithmetic, not a measure.
μ_2	$\Sigma f (x - \bar{x})^2 / N$	The variance. Its square root is the standard deviation, the standard measure of dispersion.
μ_3	$\Sigma f (x - \bar{x})^3 / N$	Carries the sign of imbalance. Cubing preserves sign, so a long right tail contributes large positive cubes and a long left tail contributes large negative cubes. Zero for a perfectly symmetric distribution.
μ_4	$\Sigma f (x - \bar{x})^4 / N$	Always positive. Fourth powers magnify distant values, so μ_4 is dominated by the peak and the tails and is the basis of kurtosis.

Note that $\mu_1 = 0$ always. This single fact simplifies every conversion formula below and is also a useful classroom check: if the signed deviations from a computed mean do not sum to zero (apart from rounding), the mean is wrong.

4. Relations between raw and central moments

Central moments can be obtained from raw moments about any point A without returning to the original data. Writing $d = x - A$ and noting that $(x - \bar{x}) = d - \mu'_1$, expansion by the binomial theorem gives the standard conversion relations:

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2(\mu'_1)^3$$

$$\mu_4 = \mu'_4 - 4\mu'_3\mu'_1 + 6\mu'_2(\mu'_1)^2 - 3(\mu'_1)^4$$

The first relation is the familiar computational formula for the variance: the mean of the squares minus the square of the mean. If the arbitrary point is chosen at the mean itself, then $\mu'_1 = 0$ and every raw moment coincides with the corresponding central moment, which confirms that the formulae are consistent. In hand computation the usual routine is therefore: choose A, build the table of powers of $(x - A)$, compute μ'_1 to μ'_4, then convert to μ_2, μ_3 and μ_4.

Sheppard corrections. When moments are computed from grouped data with class width h, grouping itself slightly inflates the even central moments. Sheppard corrections adjust for this grouping error, for example by subtracting $h^2/12$ from the grouped μ_2. The corrections assume a smooth underlying distribution tapering at both ends. They are mentioned here for completeness; the worked examples in this sheet use ungrouped data, where no correction applies.

5. Two datasets computed in full

The contrast between symmetry and skewness is clearest in two small datasets carried through the same calculation.

Dataset A (symmetric): 2, 4, 6, 8, 10. The mean is 6. The deviations from the mean are −4, −2, 0, 2, 4, perfectly balanced in sign and size.

$$\mu_2 = (16 + 4 + 0 + 4 + 16) / 5 = 8$$

$$\mu_3 = (-64 - 8 + 0 + 8 + 64) / 5 = 0$$

$$\mu_4 = (256 + 16 + 0 + 16 + 256) / 5 = 108.8$$

The third central moment vanishes exactly, as it must for any symmetric dataset: every positive cube is cancelled by an equal negative cube. From these moments, $\beta_2 = \mu_4 / \mu_2^2 = 108.8 / 64 = 1.70$, and $\beta_1 = 0$, a result used again in Sections 6 and 7.

Dataset B (skewed): 1, 2, 3, 4, 10. The mean is 4. The deviations are −3, −2, −1, 0, 6. Four values cluster below and near the mean, while the single value 10 sits far to the right and pulls the mean upward.

$$\mu_2 = (9 + 4 + 1 + 0 + 36) / 5 = 50 / 5 = 10$$

$$\mu_3 = (-27 - 8 - 1 + 0 + 216) / 5 = 180 / 5 = 36$$

$$\mu_4 = (81 + 16 + 1 + 0 + 1296) / 5 = 1394 / 5 = 278.8$$

WORKED EXAMPLES

Dataset A: mean 6; $\mu_2 = 8$; $\mu_3 = 0$; $\mu_4 = 108.8$; $\beta_2 = 108.8 / 64 =$ **1.70**; $\beta_1 =$ **0**.

Dataset B: mean 4; $\mu_2 = 10$; $\mu_3 = 36$; $\mu_4 = 278.8$; $\beta_1 = \mu_3^2 / \mu_2^3 = 1296 / 1000 =$ **1.296**; $\gamma_1 = \sqrt{\beta_1} =$ **1.139**; $\beta_2 = 278.8 / 100 =$ **2.788**.

Comparing the two datasets shows the diagnostic power of the third moment. Dataset A and dataset B have similar spreads ($\mu_2 = 8$ and 10), yet μ_3 separates them completely: 0 against 36. The positive cube of the single large deviation, $6^3 = 216$, overwhelms the small negative cubes in dataset B. That is exactly the behaviour wanted from a skewness measure: it should respond to a long tail on one side only.

6. Skewness: meaning and moment measure

Skewness is lack of symmetry. In a symmetric distribution the mean, median and mode coincide, and the two halves of the curve are mirror images. In a **positively skewed** (right-skewed) distribution a long tail stretches to the right; the mean is pulled furthest toward the tail, the median less so, and the mode stays at the peak, giving mean > median > mode. In a **negatively skewed** (left-skewed) distribution the tail stretches to the left and the order reverses: mean < median < mode. Everyday examples help: yarn strength readings with a few unusually weak specimens are often left-skewed, while counts of defects per piece, with many zeros and a few large counts, are typically right-skewed.

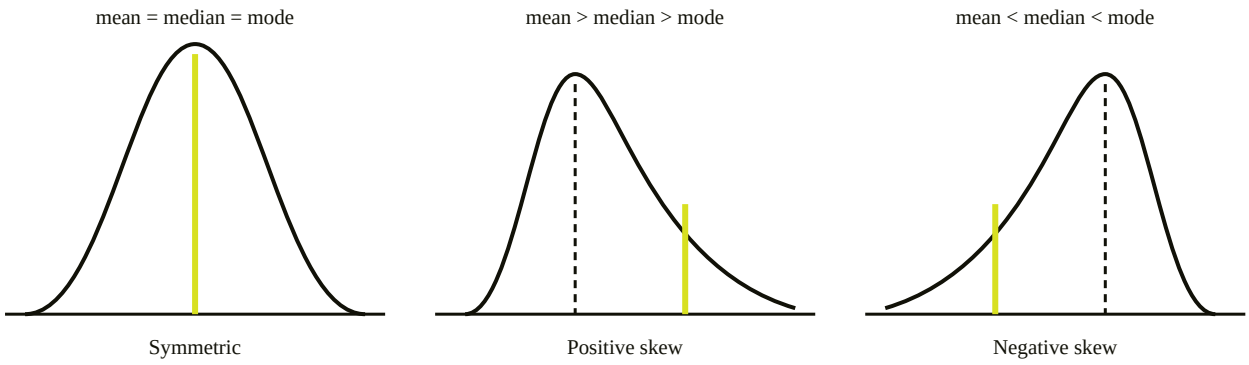


Figure 1 — Symmetric, positively skewed and negatively skewed curves, with the relative positions of mean, median and mode

A measure of skewness should be a pure number, free of units, so that distributions measured in different units can be compared. Dividing μ_3 by a suitable power of the standard deviation achieves this. Karl Pearson introduced the moment coefficients

$$\beta_1 = \mu_3^2 / \mu_2^3 , \quad \gamma_1 = \sqrt{\beta_1} = \mu_3 / \mu_2^{3/2}$$

Because μ_3 is squared in β_1, β_1 itself carries no sign; the sign is restored in γ_1, which takes the sign of μ_3. For dataset A, $\beta_1 = 0$ and $\gamma_1 = 0$, confirming symmetry. For dataset B, $\beta_1 = 1.296$ and $\gamma_1 = 1.139$, a clearly positive value signalling strong right skewness, in agreement with the picture of the data. As a rough guide, $|\gamma_1|$ below about 0.5 is mild, between 0.5 and 1 is moderate, and above 1 is strong skewness.

7. Pearson and Bowley coefficients of skewness

Moment coefficients need the full dataset, or at least its moments. Two older coefficients, still widely used in examinations, need only the familiar averages and quartiles.

Karl Pearson coefficients. Pearson reasoned that in a moderately skewed distribution the gap between the mean and the mode measures the tilt, and dividing by the standard deviation makes that gap unit-free. The **first coefficient** is

$$S_{k1} = (\text{Mean} - \text{Mode}) / \text{SD}$$

The mode of grouped data can be unstable, so Pearson also gave a **second coefficient** based on the empirical relation Mode ≈ 3 Median − 2 Mean, valid for moderately skewed unimodal distributions:

$$S_{k2} = 3\,(\text{Mean} - \text{Median}) / \text{SD}$$

Both coefficients usually lie between −1 and +1 for the first, and −3 and +3 for the second before the factor is absorbed in the conventional reporting; in practice values near zero mean near symmetry. For

the grouped marks data used across this course, the mean is 33.0, the median is 33.21, the mode is 34.0 and the standard deviation is 13.60. Hence

$$S_{k1} = (33.0 - 34.0) / 13.60 = -1 / 13.60 = -0.074$$

$$S_{k2} = 3\,(33.0 - 33.21) / 13.60 = 3\,(-0.21) / 13.60 = -0.046$$

WORKED EXAMPLE

Grouped marks data: mean 33.0, median 33.21, mode 34.0, SD 13.60.
Pearson 1st coefficient = (mean − mode) / SD = −1 / 13.60 = **−0.074**.
Pearson 2nd coefficient = 3 (mean − median) / SD = 3 (−0.21) / 13.60 = **−0.046**.
Both are small and negative: the marks distribution is very slightly negatively skewed, which is consistent because the mode (34.0) is the largest of the three averages and the mean is the smallest.

Bowley coefficient. Bowley built a coefficient entirely from quartiles, which makes it robust: extreme values cannot move the quartiles much, so the coefficient reflects the central body of the data. With lower quartile Q_1, median M and upper quartile Q_3,

$$S_B = (Q_3 + Q_1 - 2\,\text{Median}) / (Q_3 - Q_1)$$

In a symmetric distribution the median sits midway between the quartiles, the numerator vanishes, and $S_B = 0$. The coefficient always lies between −1 and +1. For the same marks data, $Q_1 = 23.25$, $Q_3 = 42.19$ and the median is 33.21, so

$$S_B = (42.19 + 23.25 - 66.42) / (42.19 - 23.25) = (-0.98) / 18.94 = -0.052$$

WORKED EXAMPLE

Bowley coefficient with $Q_1 = 23.25$, $Q_3 = 42.19$, median 33.21: $(Q_3 + Q_1 - 2\,\text{Median}) / (Q_3 - Q_1)$ = (42.19 + 23.25 − 66.42) / 18.94 = (−0.98) / 18.94 = **−0.052**. The three independent coefficients, −0.074, −0.046 and −0.052, agree closely: slight negative skewness.

The agreement of three differently built coefficients is reassuring. When coefficients disagree in sign, the moment coefficient is usually trusted for the whole distribution, while Bowley describes only the middle 50 percent. Reporting which coefficient was used is part of honest statistical practice.

8. Kurtosis

Kurtosis describes the peakedness of a distribution and, more precisely in modern usage, the weight of its tails relative to the normal curve. Its moment measure is

$$\beta_2 = \mu_4 / \mu_2^2 , \quad \gamma_2 = \beta_2 - 3$$

The normal distribution has $\beta_2 = 3$ exactly, so 3 is the benchmark. A distribution with $\beta_2 = 3$ (that is, $\gamma_2 = 0$) is **mesokurtic**, like the normal curve. A distribution with $\beta_2 > 3$ ($\gamma_2 > 0$) is **leptokurtic**: a sharper central peak with heavier tails, meaning more extreme values than the normal curve would produce. A distribution with $\beta_2 < 3$ ($\gamma_2 < 0$) is **platykurtic**: flatter at the top with lighter tails. The quantity γ_2 is called the **excess kurtosis** because it measures kurtosis in excess of the normal value.

TYPE	CONDITION	SHAPE
Mesokurtic	$\beta_2 = 3$, $\gamma_2 = 0$	Medium peak, medium tails; the normal curve is the standard example.
Leptokurtic	$\beta_2 > 3$, $\gamma_2 > 0$	Sharp peak, heavy tails; extreme values occur more often than under normality.
Platykurtic	$\beta_2 < 3$, $\gamma_2 < 0$	Flat top, light tails; values are spread more evenly with few extremes.

Applying the measure to the two datasets of Section 5: dataset A has $\beta_2 = 1.70$, well below 3, so it is platykurtic, which fits its evenly spread values with no central clustering. Dataset B has $\beta_2 = 2.788$, so $\gamma_2 = \beta_2 - 3 = -0.212$: nearly mesokurtic, slightly platykurtic. This is an instructive pairing. Dataset B is strongly skewed ($\gamma_1 = 1.139$) yet almost normal in kurtosis, proving that skewness and kurtosis capture genuinely different aspects of shape; a complete shape description needs both. In quality work the distinction matters practically: heavy tails (leptokurtosis) warn of occasional extreme fabric faults even when the average process level looks stable, while skewness warns that the process is pressing against a specification limit on one side only.

Glossary

TERM	MEANING
Raw moment	The mean of a power of deviations from an arbitrary point A, written μ'_r.
Central moment	The mean of a power of deviations from the arithmetic mean, written μ_r.
Skewness	Lack of symmetry in a distribution; the degree and direction of its lopsidedness.
β_1 and γ_1	Moment measures of skewness: $\beta_1 = \mu_3^2 / \mu_2^3$, and $\gamma_1 = \sqrt{\beta_1}$ with the sign of μ_3.

Pearson coefficient	Skewness from the averages: (mean − mode) / SD, or 3 (mean − median) / SD.
Bowley coefficient	Quartile-based skewness: $(Q_3 + Q_1 - 2\,\text{median}) / (Q_3 - Q_1)$.
Kurtosis	Peakedness and tail weight of a distribution relative to the normal curve.
β_2 and γ_2	$\beta_2 = \mu_4 / \mu_2^2$; $\gamma_2 = \beta_2 - 3$ is the excess kurtosis.
Mesokurtic, leptokurtic, platykurtic	Kurtosis equal to, greater than, or less than the normal value $\beta_2 = 3$, respectively.

Key points & summary

- Raw moments are taken about an arbitrary point A for easy computation; central moments are taken about the mean and describe shape.
- $\mu_1 = 0$ always; μ_2 is the variance, μ_3 carries skewness, and μ_4 carries kurtosis.
- Conversion: $\mu_2 = \mu'_2 - (\mu'_1)^2$, with parallel formulae for μ_3 and μ_4.
- Dataset A (2, 4, 6, 8, 10) gives $\mu_3 = 0$ and $\beta_2 = 1.70$; dataset B (1, 2, 3, 4, 10) gives $\mu_3 = 36$, $\gamma_1 = 1.139$ and $\beta_2 = 2.788$.
- Positive skew means mean > median > mode; negative skew reverses the order.
- For the marks data the Pearson coefficients are −0.074 and −0.046 and Bowley gives −0.052: all show slight negative skewness.
- Kurtosis is judged against $\beta_2 = 3$: equal is mesokurtic, greater is leptokurtic, smaller is platykurtic.

Practice

Answer in your own words; brief model answers follow.

1. Define raw and central moments, and explain why μ_1 is always zero.
2. For the data 2, 4, 6, 8, 10, verify that $\mu_2 = 8$ and $\mu_4 = 108.8$, and state the value of β_2 and what it says about the shape.
3. For the data 1, 2, 3, 4, 10, show how $\beta_1 = 1.296$ and $\gamma_1 = 1.139$ are obtained, and interpret the sign.
4. Using mean 33.0, median 33.21, mode 34.0 and SD 13.60, compute both Pearson coefficients of skewness and comment on the result.
5. Using $Q_1 = 23.25$, $Q_3 = 42.19$ and median 33.21, compute the Bowley coefficient and explain one advantage it has over the Pearson coefficients.

6. Distinguish mesokurtic, leptokurtic and platykurtic distributions in terms of β_2, and classify a distribution with $\beta_2 = 2.788$.

Answer key

1. A raw moment μ'_r is the mean of the r-th power of deviations from an arbitrary point A; a central moment μ_r uses deviations from the mean. $\mu_1 = \Sigma(x - \bar{x}) / N = 0$ because deviations from the mean sum to zero by the defining property of the mean.
2. Deviations are −4, −2, 0, 2, 4. Squares sum to 40, so $\mu_2 = 40 / 5 = 8$. Fourth powers sum to 544, so $\mu_4 = 544 / 5 = 108.8$. $\beta_2 = 108.8 / 64 = 1.70 < 3$, so the distribution is platykurtic.
3. $\mu_2 = 10$ and $\mu_3 = 36$, so $\beta_1 = 36^2 / 10^3 = 1296 / 1000 = 1.296$ and $\gamma_1 = \sqrt{1.296} = 1.139$. The positive sign, inherited from $\mu_3 > 0$, means strong positive (right) skewness, caused by the single large value 10.
4. First coefficient: (33.0 − 34.0) / 13.60 = −0.074. Second coefficient: 3 (33.0 − 33.21) / 13.60 = −0.046. Both are close to zero and negative, so the distribution is almost symmetric with a very slight negative skew.
5. $S_B = (42.19 + 23.25 - 66.42) / (42.19 - 23.25) = (-0.98) / 18.94 = -0.052$. Advantage: it uses only quartiles, so it is unaffected by a few extreme values that can distort the mean, mode and standard deviation.
6. Mesokurtic: $\beta_2 = 3$. Leptokurtic: $\beta_2 > 3$, sharp peak and heavy tails. Platykurtic: $\beta_2 < 3$, flat top and light tails. $\beta_2 = 2.788$ is below 3 with $\gamma_2 = -0.212$, so the distribution is nearly mesokurtic, slightly platykurtic.

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